

Vendor & Technology Assessment

Oracle, SAP, AWS, and open-source evaluation

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1. Introduction

This document assesses the technology vendors and platforms used by ArchiSurance, with a focus on strategic alignment, total cost of ownership, and risk. It informs the Cloud Migration Assessment (EA-CLD-009) and the Strategic IT Roadmap (EA-ROAD-012).

The incident management process defines four severity levels, with Severity 1 incidents requiring a response within fifteen minutes and a post-incident review within five business days. The team uses a definition of done that includes code review by at least two peers, passing CI pipeline, updated architecture diagrams, and entry in the change log maintained by the Release Manager. Automated deployment pipelines leverage the CI/CD infrastructure described in the platform engineering documentation. The enterprise backup strategy uses a 3-2-1 approach: three copies of data, on two different media types, with one copy stored offsite at the secondary data center in Rotterdam. The design considers both current-state constraints and target-state aspirations as documented in the strategic roadmap. Dependencies on third-party services were documented and risk-assessed, with fallback mechanisms defined for each critical integration point. Capacity planning exercises revealed that the current infrastructure can sustain projected growth for the next twelve months, after which additional compute and storage resources will be required. Feedback from the user acceptance testing phase indicated that the new interface reduced the average task completion time by approximately thirty percent compared to the legacy system, though some users requested additional keyboard shortcuts for power-user workflows. Monitoring and alerting are integrated with the central observability stack based on Prometheus and Grafana. Performance benchmarks are captured in the enterprise monitoring framework and reported on a monthly basis to leadership. Security review was completed in accordance with the CISO's security-by-design framework. Cross-functional workshops held in January and March 2025 helped refine the scope and integration points. Post-implementation review sessions are scheduled for thirty and ninety days after go-live to capture lessons learned and identify optimization opportunities. Cost allocation for shared infrastructure follows a chargeback model based on resource consumption metrics, with monthly reports sent to each business unit for transparency and accountability. The platform engineering team provides a standardized deployment template that all application teams are required to use, ensuring consistency in logging, health checks, and resource limits. Stakeholders across multiple departments have contributed requirements for this workstream. The enterprise data model, maintained by the Data Architecture team, defines canonical schemas for all shared data objects; integrating systems are expected to conform to these schemas or implement transformation layers that map proprietary formats to the canonical model. Business continuity considerations were incorporated from the outset, with the DR Recovery Node in the secondary data center providing failover capacity; the most recent failover test was conducted in October 2024 and achieved recovery within the target RTO of one hour. Key performance indicators for the initiative include transaction throughput, error rate, mean time to detection, mean time to recovery, and user satisfaction scores collected through quarterly surveys. Budget allocation was approved as part of the annual IT investment portfolio review conducted in November 2024. The business process documentation follows the BPMN 2.0 standard, with process models maintained in the Signavio process modeling tool and linked to the ArchiMate model elements in Sparx EA. The project governance structure includes a steering committee meeting monthly, a working group meeting weekly, and daily stand-ups during active development sprints. The operational acceptance criteria include automated smoke tests, performance baselines, runbook completeness, and sign-off from both the IT Service Manager and the business process owner.

Collaboration with the vendor support team has been essential during the implementation phase; weekly technical syncs ensure that configuration changes are validated against the vendor's reference architecture before being applied to the production environment. The solution was designed to be modular, allowing individual components to be upgraded or replaced without affecting the overall system availability or data integrity. ArchiSurance participates in an industry-wide information sharing and analysis center for insurance companies, contributing anonymized threat intelligence data and receiving early warnings about sector-specific attack patterns. A comprehensive training program was developed for end users, including online modules, instructor-led sessions, and a sandbox environment for hands-on practice. The incident management process defines four severity levels, with Severity 1 incidents requiring a response within fifteen minutes and a post-incident review within five business days. The API design guidelines specify that all REST endpoints must support content negotiation, return standard HTTP status codes, include correlation identifiers in response headers, and provide machine-readable error bodies in RFC 7807 format. The data retention policy mandates that transactional records are kept for seven years in online storage and archived to cold storage thereafter, with the ability to retrieve

archived records within twenty-four hours upon regulatory request. Service-level indicators are collected continuously and aggregated into dashboards visible to both technical teams and business stakeholders.

Deployment windows are scheduled every Tuesday and Thursday between 06:00 and 08:00 CET, with emergency deployments requiring approval from both the IT Service Manager and the on-call engineer. Capacity forecasting models project a twenty-five percent year-over-year increase in data volume, driven primarily by the digitization of paper-based processes and the expansion of the broker channel. Automated deployment pipelines leverage the CI/CD infrastructure described in the platform engineering documentation. The architecture decisions were reviewed and approved by the Architecture Review Board in Q2 2024. The architecture team conducted a threat modeling exercise using the STRIDE framework, identifying potential spoofing, tampering, and information disclosure risks; mitigations for each threat were documented and tracked in the security risk register. The enterprise identity provider federates with the national eHerkenning system for business-to-government interactions and with DigiD for customer-facing authentication on the self-service portal. The network team conducted a bandwidth utilization study in February 2025, confirming that inter-datacenter traffic remains below sixty percent of the available WAN capacity under normal operating conditions. The design considers both current-state constraints and target-state aspirations as documented in the strategic roadmap.

The architecture team maintains a dependency register that maps each component to its upstream providers and downstream consumers, updated quarterly. The Architecture Review Board convened a special session in February 2025 to evaluate the implications of the proposed changes on downstream systems and business processes. Budget allocation was approved as part of the annual IT investment portfolio review conducted in November 2024. The platform engineering team provides a standardized deployment template that all application teams are required to use, ensuring consistency in logging, health checks, and resource limits. Capacity planning exercises revealed that the current infrastructure can sustain projected growth for the next twelve months, after which additional compute and storage resources will be required. The data classification exercise, completed in Q4 2024, assigned sensitivity labels to all data objects, informing encryption, access control, and retention policies. Cross-functional workshops held in January and March 2025 helped refine the scope and integration points. An accessibility audit was conducted by an external specialist in January 2025 to ensure that all user-facing components meet WCAG 2.1 Level AA standards, with remediation items tracked in the product backlog.

2. Database Technology Assessment

PostgreSQL 15

Serves: Policy Administration System, Claims Processing System, Customer Management Platform.

Compliance with the General Data Protection Regulation requires that personally identifiable information is encrypted at rest and in transit, with access restricted to authorized roles as defined in the Identity and Access Management policy. The initiative is tracked in the enterprise PMO tool with bi-weekly status updates to the steering committee. The architecture decisions were reviewed and approved by the Architecture Review Board in Q2 2024. A dedicated Slack channel was established for real-time coordination between development, operations, and security teams during the rollout period. Interoperability with partner systems is achieved through the Partner Integration Hub, which exposes a curated subset of the internal API surface with additional rate limiting, schema validation, and audit logging applied at the gateway layer. Dependencies on third-party services were documented and risk-assessed, with fallback mechanisms defined for each critical integration point. The architecture team conducted a threat modeling exercise using the STRIDE framework, identifying potential spoofing, tampering, and information disclosure risks; mitigations for each threat were documented and tracked in the security risk register. In the context of insurance operations, peak load occurs during natural disaster events when claim submissions can spike by up to ten times the normal rate; the auto-scaling configuration accounts for this burst pattern. External consultants from Deloitte provided an independent assessment of the architecture in late 2024. From a historical perspective, ArchiSurance first recognized the need for modernization in this area following an internal efficiency review conducted in late 2022, which highlighted significant manual effort and process duplication across regional offices. A chaos engineering program was introduced in Q1 2025, starting with controlled fault injection in the staging environment to validate the resilience of the system against network

partitions and dependency failures. ArchiSurance participates in an industry-wide information sharing and analysis center for insurance companies, contributing anonymized threat intelligence data and receiving early warnings about sector-specific attack patterns. Lessons learned from the pilot phase have been incorporated into the current design iteration. The total cost of ownership analysis, prepared by the Finance Department in collaboration with IT, indicated a payback period of approximately eighteen months assuming current utilization rates. Capacity planning exercises revealed that the current infrastructure can sustain projected growth for the next twelve months, after which additional compute and storage resources will be required. The development team follows trunk-based development with short-lived feature branches, automated linting, and a code coverage threshold of eighty percent enforced by the CI pipeline. The observability strategy follows the three pillars model: structured logs shipped to Elasticsearch, distributed traces collected via OpenTelemetry, and metrics scraped by Prometheus; all three are unified in Grafana dashboards. An internal audit in March 2025 confirmed that the controls implemented for this area meet the requirements of the enterprise risk management framework. An accessibility audit was conducted by an external specialist in January 2025 to ensure that all user-facing components meet WCAG 2.1 Level AA standards, with remediation items tracked in the product backlog. The solution leverages event-driven patterns wherever possible, with Apache Kafka providing durable, ordered event streams that decouple producers from consumers and enable replay for debugging and reprocessing scenarios. Stakeholders across multiple departments have contributed requirements for this workstream. Cost allocation for shared infrastructure follows a chargeback model based on resource consumption metrics, with monthly reports sent to each business unit for transparency and accountability. Integration testing spans three environments: development, staging, and pre-production. Service-level indicators are collected continuously and aggregated into dashboards visible to both technical teams and business stakeholders. Monitoring and alerting are integrated with the central observability stack based on Prometheus and Grafana. The Architecture Review Board convened a special session in February 2025 to evaluate the implications of the proposed changes on downstream systems and business processes. Post-implementation review sessions are scheduled for thirty and ninety days after go-live to capture lessons learned and identify optimization opportunities. A formal lessons-learned workshop is held at the end of each quarter, with outcomes published to the internal wiki and action items assigned to specific team members with target completion dates.

Oracle DB

Serves: Billing System, Enterprise Data Warehouse.

The solution leverages event-driven patterns wherever possible, with Apache Kafka providing durable, ordered event streams that decouple producers from consumers and enable replay for debugging and reprocessing scenarios. The enterprise backup strategy uses a 3-2-1 approach: three copies of data, on two different media types, with one copy stored offsite at the secondary data center in Rotterdam. The enterprise architecture repository in Sparx EA is the authoritative source for all model elements; any changes to this document must be reflected there within five business days. The operational acceptance criteria include automated smoke tests, performance baselines, runbook completeness, and sign-off from both the IT Service Manager and the business process owner. The development team follows trunk-based development with short-lived feature branches, automated linting, and a code coverage threshold of eighty percent enforced by the CI pipeline. An internal audit in March 2025 confirmed that the controls implemented for this area meet the requirements of the enterprise risk management framework. Budget allocation was approved as part of the annual IT investment portfolio review conducted in November 2024. The architecture team maintains a dependency register that maps each component to its upstream providers and downstream consumers, updated quarterly. Interoperability with partner systems is achieved through the Partner Integration Hub, which exposes a curated subset of the internal API surface with additional rate limiting, schema validation, and audit logging applied at the gateway layer. Implementation follows an iterative approach, with quarterly milestones aligned to the overall program timeline. The enterprise identity provider federates with the national eHerkenning system for business-to-government interactions and with DigiD for customer-facing authentication on the self-service portal. The project governance structure includes a steering committee meeting monthly, a working group meeting weekly, and daily stand-ups during active development sprints. Monitoring and alerting are integrated with the central observability stack based on Prometheus and Grafana. The enterprise data model, maintained by the Data Architecture team, defines canonical schemas for all shared data objects; integrating systems are expected to conform to these schemas or implement transformation layers that map proprietary formats to the canonical model. The migration strategy follows a strangler-fig pattern, where new functionality is built in the target platform while legacy features are gradually redirected; this

approach minimizes disruption to active users and allows incremental validation of the new implementation. The vendor selection process followed the procurement guidelines established by the CFO's office. Lessons learned from the pilot phase have been incorporated into the current design iteration. Compliance with the General Data Protection Regulation requires that personally identifiable information is encrypted at rest and in transit, with access restricted to authorized roles as defined in the Identity and Access Management policy. The transition plan was developed in alignment with the enterprise change management framework, incorporating staged rollouts, parallel-run periods, and defined go/no-go criteria at each milestone. Service-level indicators are collected continuously and aggregated into dashboards visible to both technical teams and business stakeholders. The data retention policy mandates that transactional records are kept for seven years in online storage and archived to cold storage thereafter, with the ability to retrieve archived records within twenty-four hours upon regulatory request. Regression test suites are maintained by the QA team and executed automatically on every merge to the main branch via the CI/CD pipeline. Documentation standards require that all architecture decisions are recorded as Architecture Decision Records using the ADR format proposed by Michael Nygard, stored alongside the codebase in the version control system. From a historical perspective, ArchiSurance first recognized the need for modernization in this area following an internal efficiency review conducted in late 2022, which highlighted significant manual effort and process duplication across regional offices. An accessibility audit was conducted by an external specialist in January 2025 to ensure that all user-facing components meet WCAG 2.1 Level AA standards, with remediation items tracked in the product backlog. Security review was completed in accordance with the CISO's security-by-design framework. The solution was designed to be modular, allowing individual components to be upgraded or replaced without affecting the overall system availability or data integrity. Integration testing spans three environments: development, staging, and pre-production.

Microsoft SQL Server 2022

Serves: BI Platform.

Regression test suites are maintained by the QA team and executed automatically on every merge to the main branch via the CI/CD pipeline. Cost allocation for shared infrastructure follows a chargeback model based on resource consumption metrics, with monthly reports sent to each business unit for transparency and accountability. Collaboration with the vendor support team has been essential during the implementation phase; weekly technical syncs ensure that configuration changes are validated against the vendor's reference architecture before being applied to the production environment. Scalability testing performed in the staging environment confirmed that the system handles a three-times-peak load without degradation, providing sufficient headroom for seasonal demand fluctuations. The API design guidelines specify that all REST endpoints must support content negotiation, return standard HTTP status codes, include correlation identifiers in response headers, and provide machine-readable error bodies in RFC 7807 format. Data privacy impact assessment was filed with the Data Protection Officer in Q1 2025. The solution was designed to be modular, allowing individual components to be upgraded or replaced without affecting the overall system availability or data integrity. The vendor selection process followed the procurement guidelines established by the CFO's office. Application health is assessed using a synthetic monitoring approach, where automated scripts simulate real user journeys every five minutes and report availability from three geographically distributed probe locations. Capacity planning exercises revealed that the current infrastructure can sustain projected growth for the next twelve months, after which additional compute and storage resources will be required. A formal lessons-learned workshop is held at the end of each quarter, with outcomes published to the internal wiki and action items assigned to specific team members with target completion dates. The enterprise backup strategy uses a 3-2-1 approach: three copies of data, on two different media types, with one copy stored offsite at the secondary data center in Rotterdam. The network team conducted a bandwidth utilization study in February 2025, confirming that inter-datacenter traffic remains below sixty percent of the available WAN capacity under normal operating conditions. The business process documentation follows the BPMN 2.0 standard, with process models maintained in the Signavio process modeling tool and linked to the ArchiMate model elements in Sparx EA. Documentation standards require that all architecture decisions are recorded as Architecture Decision Records using the ADR format proposed by Michael Nygard, stored alongside the codebase in the version control system. Automated deployment pipelines leverage the CI/CD infrastructure described in the platform engineering documentation. During the initial assessment phase, the team identified several areas where the existing architecture fell short of the target-state requirements defined in the enterprise roadmap. The architecture review confirmed that the proposed solution aligns with the TOGAF Architecture Development Method phases used by ArchiSurance, specifically phases B through D covering Business, Information Systems, and Technology Architecture. Risk mitigation strategies

include regular checkpoint reviews, automated regression testing, and staged rollout procedures. An internal audit in March 2025 confirmed that the controls implemented for this area meet the requirements of the enterprise risk management framework. The data classification exercise, completed in Q4 2024, assigned sensitivity labels to all data objects, informing encryption, access control, and retention policies. Compliance with the General Data Protection Regulation requires that personally identifiable information is encrypted at rest and in transit, with access restricted to authorized roles as defined in the Identity and Access Management policy. Budget allocation was approved as part of the annual IT investment portfolio review conducted in November 2024. Lessons learned from the pilot phase have been incorporated into the current design iteration.

3. Container Platform Assessment

the container orchestration platform

Hosts: Claims Platform, Risk Engine, Anti-Fraud Module, Gateway, Messaging Service.

The team uses a definition of done that includes code review by at least two peers, passing CI pipeline, updated architecture diagrams, and entry in the change log maintained by the Release Manager. An internal audit in March 2025 confirmed that the controls implemented for this area meet the requirements of the enterprise risk management framework. Monitoring and alerting are integrated with the central observability stack based on Prometheus and Grafana. The transition plan was developed in alignment with the enterprise change management framework, incorporating staged rollouts, parallel-run periods, and defined go/no-go criteria at each milestone. The initiative is tracked in the enterprise PMO tool with bi-weekly status updates to the steering committee. Data privacy impact assessment was filed with the Data Protection Officer in Q1 2025. The total cost of ownership analysis, prepared by the Finance Department in collaboration with IT, indicated a payback period of approximately eighteen months assuming current utilization rates. Implementation follows an iterative approach, with quarterly milestones aligned to the overall program timeline. The data classification exercise, completed in Q4 2024, assigned sensitivity labels to all data objects, informing encryption, access control, and retention policies. Dependencies on third-party services were documented and risk-assessed, with fallback mechanisms defined for each critical integration point. Deployment windows are scheduled every Tuesday and Thursday between 06:00 and 08:00 CET, with emergency deployments requiring approval from both the IT Service Manager and the on-call engineer. The API design guidelines specify that all REST endpoints must support content negotiation, return standard HTTP status codes, include correlation identifiers in response headers, and provide machine-readable error bodies in RFC 7807 format. A formal lessons-learned workshop is held at the end of each quarter, with outcomes published to the internal wiki and action items assigned to specific team members with target completion dates. The solution leverages event-driven patterns wherever possible, with Apache Kafka providing durable, ordered event streams that decouple producers from consumers and enable replay for debugging and reprocessing scenarios. Performance benchmarks are captured in the enterprise monitoring framework and reported on a monthly basis to leadership. Capacity forecasting models project a twenty-five percent year-over-year increase in data volume, driven primarily by the digitization of paper-based processes and the expansion of the broker channel. The vendor selection process followed the procurement guidelines established by the CFO's office. The architecture team maintains a dependency register that maps each component to its upstream providers and downstream consumers, updated quarterly. The non-functional requirements specify a maximum end-to-end latency of five hundred milliseconds at the ninety-ninth percentile, measured under a sustained load of one thousand concurrent users. The enterprise architecture repository in Sparx EA is the authoritative source for all model elements; any changes to this document must be reflected there within five business days. The operational acceptance criteria include automated smoke tests, performance baselines, runbook completeness, and sign-off from both the IT Service Manager and the business process owner. The data retention policy mandates that transactional records are kept for seven years in online storage and archived to cold storage thereafter, with the ability to retrieve archived records within twenty-four hours upon regulatory request.

the container runtime

Hosts: Salesforce CRM, Broker Portal, Mobile App Backend.

A chaos engineering program was introduced in Q1 2025, starting with controlled fault injection in the staging environment to validate the resilience of the system against network partitions and dependency failures. The information security team requires that all new systems undergo a penetration test before production deployment, with findings above medium severity remediated prior to go-live approval. The vendor selection process followed the procurement guidelines established by the CFO's office. During the initial assessment phase, the team identified several areas where the existing architecture fell short of the target-state requirements defined in the enterprise roadmap. The transition plan was developed in alignment with the enterprise change management framework, incorporating staged rollouts, parallel-run periods, and defined go/no-go criteria at each milestone. The project governance structure includes a steering committee meeting monthly, a working group meeting weekly, and daily stand-ups during active development sprints. Risk mitigation strategies include regular checkpoint reviews, automated regression testing, and staged rollout procedures. The enterprise technology radar, published semi-annually by the CTO's office, classifies technologies as Adopt, Trial, Assess, or Hold; the technologies used in this area are all currently in the Adopt or Trial rings. Stakeholders across multiple departments have contributed requirements for this workstream. The data classification exercise, completed in Q4 2024, assigned sensitivity labels to all data objects, informing encryption, access control, and retention policies. A formal lessons-learned workshop is held at the end of each quarter, with outcomes published to the internal wiki and action items assigned to specific team members with target completion dates. The enterprise identity provider federates with the national eHerkenning system for business-to-government interactions and with DigiD for customer-facing authentication on the self-service portal. Capacity planning exercises revealed that the current infrastructure can sustain projected growth for the next twelve months, after which additional compute and storage resources will be required. The solution was designed to be modular, allowing individual components to be upgraded or replaced without affecting the overall system availability or data integrity. Change management activities include stakeholder communication, training sessions, and documentation updates. A proof-of-concept was conducted over a four-week period, involving representatives from the Claims Department, IT Operations, and the Data Analytics Team to validate the technical feasibility. The team uses a definition of done that includes code review by at least two peers, passing CI pipeline, updated architecture diagrams, and entry in the change log maintained by the Release Manager. Documentation standards require that all architecture decisions are recorded as Architecture Decision Records using the ADR format proposed by Michael Nygard, stored alongside the codebase in the version control system. Regression test suites are maintained by the QA team and executed automatically on every merge to the main branch via the CI/CD pipeline. Post-implementation review sessions are scheduled for thirty and ninety days after go-live to capture lessons learned and identify optimization opportunities. Business continuity considerations were incorporated from the outset, with the DR Recovery Node in the secondary data center providing failover capacity; the most recent failover test was conducted in October 2024 and achieved recovery within the target RTO of one hour. The configuration management database maintains a record of all IT assets, including hardware serial numbers, software license keys, patch levels, and relationships between configuration items. The enterprise backup strategy uses a 3-2-1 approach: three copies of data, on two different media types, with one copy stored offsite at the secondary data center in Rotterdam. Dependencies on third-party services were documented and risk-assessed, with fallback mechanisms defined for each critical integration point.

4. Integration Technology Assessment

Kafka

Supports: MuleSoft ESB, Batch Processing Engine.

Key performance indicators for the initiative include transaction throughput, error rate, mean time to detection, mean time to recovery, and user satisfaction scores collected through quarterly surveys. The design considers both current-state constraints and target-state aspirations as documented in the strategic roadmap. Cost allocation for shared infrastructure follows a chargeback model based on resource consumption metrics, with monthly reports sent to each business unit for transparency and accountability. The architecture team conducted a threat modeling exercise using the STRIDE framework, identifying potential spoofing, tampering, and information disclosure risks; mitigations for each threat were documented and tracked in the security risk register.

the in-memory cache

Supports: Gateway, Customer Self-Service Portal.

The data retention policy mandates that transactional records are kept for seven years in online storage and archived to cold storage thereafter, with the ability to retrieve archived records within twenty-four hours upon regulatory request. The initiative is tracked in the enterprise PMO tool with bi-weekly status updates to the steering committee. Scalability testing performed in the staging environment confirmed that the system handles a three-times-peak load without degradation, providing sufficient headroom for seasonal demand fluctuations. The observability strategy follows the three pillars model: structured logs shipped to Elasticsearch, distributed traces collected via OpenTelemetry, and metrics scraped by Prometheus; all three are unified in Grafana dashboards.

Elasticsearch

Supports: DMS.

The data retention policy mandates that transactional records are kept for seven years in online storage and archived to cold storage thereafter, with the ability to retrieve archived records within twenty-four hours upon regulatory request. ArchiSurance participates in an industry-wide information sharing and analysis center for insurance companies, contributing anonymized threat intelligence data and receiving early warnings about sector-specific attack patterns. The migration strategy follows a strangler-fig pattern, where new functionality is built in the target platform while legacy features are gradually redirected; this approach minimizes disruption to active users and allows incremental validation of the new implementation. The design considers both current-state constraints and target-state aspirations as documented in the strategic roadmap.

Nginx

Supports: Content Delivery Network, MyArchiSurance Portal.

The data retention policy mandates that transactional records are kept for seven years in online storage and archived to cold storage thereafter, with the ability to retrieve archived records within twenty-four hours upon regulatory request. Cross-functional workshops held in January and March 2025 helped refine the scope and integration points. The transition plan was developed in alignment with the enterprise change management framework, incorporating staged rollouts, parallel-run periods, and defined go/no-go criteria at each milestone. A dedicated Slack channel was established for real-time coordination between development, operations, and security teams during the rollout period.

5. Vendor Risk Analysis

The constraint 'Vendor Lock-in Avoidance' drives the preference for open-source and cloud-agnostic technologies. The Third-Party Data Provider Contract introduces a dependency on external data quality. The Legacy Oracle DB Migration Deadline Q4 2026 creates urgency for the database migration.

Vendor/Tech	Risk Level	Lock-in	Alternative
Oracle DB 19c	High	High	PostgreSQL 15
Kubernetes	Low	Low	Nomad / ECS
Apache Kafka	Medium	Medium	RabbitMQ / Pulsar
Elasticsearch	Medium	Medium	OpenSearch
Redis	Low	Low	Memcached / Valkey
Nginx	Low	Low	HAProxy / Envoy

The migration strategy follows a strangler-fig pattern, where new functionality is built in the target platform while legacy features are gradually redirected; this approach minimizes disruption to active users and allows incremental validation of the new implementation. The enterprise architecture repository in Sparx EA is the authoritative source for all model elements; any changes to this document must be reflected there within five business days. The platform engineering

team provides a standardized deployment template that all application teams are required to use, ensuring consistency in logging, health checks, and resource limits. An accessibility audit was conducted by an external specialist in January 2025 to ensure that all user-facing components meet WCAG 2.1 Level AA standards, with remediation items tracked in the product backlog. The Architecture Review Board convened a special session in February 2025 to evaluate the implications of the proposed changes on downstream systems and business processes. The non-functional requirements specify a maximum end-to-end latency of five hundred milliseconds at the ninety-ninth percentile, measured under a sustained load of one thousand concurrent users. A dedicated Slack channel was established for real-time coordination between development, operations, and security teams during the rollout period. Cross-functional workshops held in January and March 2025 helped refine the scope and integration points. The data retention policy mandates that transactional records are kept for seven years in online storage and archived to cold storage thereafter, with the ability to retrieve archived records within twenty-four hours upon regulatory request. Automated deployment pipelines leverage the CI/CD infrastructure described in the platform engineering documentation. The configuration management database maintains a record of all IT assets, including hardware serial numbers, software license keys, patch levels, and relationships between configuration items. Scalability testing performed in the staging environment confirmed that the system handles a three-times-peak load without degradation, providing sufficient headroom for seasonal demand fluctuations. Lessons learned from the pilot phase have been incorporated into the current design iteration. The development team follows trunk-based development with short-lived feature branches, automated linting, and a code coverage threshold of eighty percent enforced by the CI pipeline. Security review was completed in accordance with the CISO's security-by-design framework. The total cost of ownership analysis, prepared by the Finance Department in collaboration with IT, indicated a payback period of approximately eighteen months assuming current utilization rates. The vendor selection process followed the procurement guidelines established by the CFO's office. Capacity planning exercises revealed that the current infrastructure can sustain projected growth for the next twelve months, after which additional compute and storage resources will be required. The solution was designed to be modular, allowing individual components to be upgraded or replaced without affecting the overall system availability or data integrity. During the initial assessment phase, the team identified several areas where the existing architecture fell short of the target-state requirements defined in the enterprise roadmap. The project governance structure includes a steering committee meeting monthly, a working group meeting weekly, and daily stand-ups during active development sprints. Stakeholders across multiple departments have contributed requirements for this workstream. The network team conducted a bandwidth utilization study in February 2025, confirming that inter-datacenter traffic remains below sixty percent of the available WAN capacity under normal operating conditions. Dependencies on third-party services were documented and risk-assessed, with fallback mechanisms defined for each critical integration point. The vendor risk assessment framework evaluates suppliers across five dimensions: financial stability, technology roadmap alignment, support quality, security posture, and contractual flexibility.

Capacity planning exercises revealed that the current infrastructure can sustain projected growth for the next twelve months, after which additional compute and storage resources will be required. The observability strategy follows the three pillars model: structured logs shipped to Elasticsearch, distributed traces collected via OpenTelemetry, and metrics scraped by Prometheus; all three are unified in Grafana dashboards. Lessons learned from the pilot phase have been incorporated into the current design iteration. The architecture team maintains a dependency register that maps each component to its upstream providers and downstream consumers, updated quarterly. The architectural principles that guide this work include separation of concerns, loose coupling between components, preference for open standards, and the ability to scale horizontally under peak load conditions such as end-of-year policy renewals. During the initial assessment phase, the team identified several areas where the existing architecture fell short of the target-state requirements defined in the enterprise roadmap. Integration testing spans three environments: development, staging, and pre-production. Cost allocation for shared infrastructure follows a chargeback model based on resource consumption metrics, with monthly reports sent to each business unit for transparency and accountability. Security review was completed in accordance with the CISO's security-by-design framework. Scalability testing performed in the staging environment confirmed that the system handles a three-times-peak load without degradation, providing sufficient headroom for seasonal demand fluctuations.

Compliance with the General Data Protection Regulation requires that personally identifiable information is encrypted at rest and in transit, with access restricted to authorized roles as defined in the Identity and Access Management policy. Performance benchmarks are captured in the enterprise monitoring framework and reported on a monthly basis to leadership. The network team conducted a bandwidth utilization study in February 2025, confirming that inter-datacenter traffic remains below sixty percent of the available WAN capacity under normal operating conditions. Interoperability with

partner systems is achieved through the Partner Integration Hub, which exposes a curated subset of the internal API surface with additional rate limiting, schema validation, and audit logging applied at the gateway layer. In the context of insurance operations, peak load occurs during natural disaster events when claim submissions can spike by up to ten times the normal rate; the auto-scaling configuration accounts for this burst pattern. During the initial assessment phase, the team identified several areas where the existing architecture fell short of the target-state requirements defined in the enterprise roadmap. The architecture team conducted a threat modeling exercise using the STRIDE framework, identifying potential spoofing, tampering, and information disclosure risks; mitigations for each threat were documented and tracked in the security risk register. Deployment windows are scheduled every Tuesday and Thursday between 06:00 and 08:00 CET, with emergency deployments requiring approval from both the IT Service Manager and the on-call engineer. The vendor risk assessment framework evaluates suppliers across five dimensions: financial stability, technology roadmap alignment, support quality, security posture, and contractual flexibility. The design considers both current-state constraints and target-state aspirations as documented in the strategic roadmap.

The Architecture Review Board convened a special session in February 2025 to evaluate the implications of the proposed changes on downstream systems and business processes. Collaboration with the vendor support team has been essential during the implementation phase; weekly technical syncs ensure that configuration changes are validated against the vendor's reference architecture before being applied to the production environment. A formal lessons-learned workshop is held at the end of each quarter, with outcomes published to the internal wiki and action items assigned to specific team members with target completion dates. Monitoring and alerting are integrated with the central observability stack based on Prometheus and Grafana. During the initial assessment phase, the team identified several areas where the existing architecture fell short of the target-state requirements defined in the enterprise roadmap. Automated deployment pipelines leverage the CI/CD infrastructure described in the platform engineering documentation. A proof-of-concept was conducted over a four-week period, involving representatives from the Claims Department, IT Operations, and the Data Analytics Team to validate the technical feasibility. Cross-functional workshops held in January and March 2025 helped refine the scope and integration points. Technical debt in this area was cataloged during a dedicated sprint in Q3 2024; the resulting backlog of forty-seven items has been prioritized by business impact and is being addressed alongside feature development. Regression test suites are maintained by the QA team and executed automatically on every merge to the main branch via the CI/CD pipeline.

6. Recommendations

Based on the assessment, the Architecture Board recommends: (1) Accelerate Oracle to PostgreSQL migration, (2) Adopt managed Kubernetes in cloud, (3) Evaluate OpenSearch as Elasticsearch alternative, (4) Maintain Redis and Nginx as they present low vendor risk.

Scalability testing performed in the staging environment confirmed that the system handles a three-times-peak load without degradation, providing sufficient headroom for seasonal demand fluctuations. The platform engineering team provides a standardized deployment template that all application teams are required to use, ensuring consistency in logging, health checks, and resource limits. The total cost of ownership analysis, prepared by the Finance Department in collaboration with IT, indicated a payback period of approximately eighteen months assuming current utilization rates. Integration testing spans three environments: development, staging, and pre-production. Data privacy impact assessment was filed with the Data Protection Officer in Q1 2025. Capacity planning exercises revealed that the current infrastructure can sustain projected growth for the next twelve months, after which additional compute and storage resources will be required. The technology vendor recommendations has been a critical area of focus for ArchiSurance since the digital transformation initiative began in 2023. The architecture team maintains a dependency register that maps each component to its upstream providers and downstream consumers, updated quarterly. Stakeholders across multiple departments have contributed requirements for this workstream. The observability strategy follows the three pillars model: structured logs shipped to Elasticsearch, distributed traces collected via OpenTelemetry, and metrics scraped by Prometheus; all three are unified in Grafana dashboards. The architecture review confirmed that the proposed solution aligns with the TOGAF Architecture Development Method phases used by ArchiSurance, specifically phases B through D covering Business, Information Systems, and Technology Architecture. The API design guidelines specify that all REST endpoints must support content negotiation, return standard HTTP status codes, include correlation identifiers in response headers, and provide machine-readable error bodies in RFC 7807 format. The vendor risk assessment framework evaluates suppliers across five dimensions: financial stability, technology roadmap

alignment, support quality, security posture, and contractual flexibility. ArchiSurance participates in an industry-wide information sharing and analysis center for insurance companies, contributing anonymized threat intelligence data and receiving early warnings about sector-specific attack patterns. The operational acceptance criteria include automated smoke tests, performance baselines, runbook completeness, and sign-off from both the IT Service Manager and the business process owner. A comprehensive training program was developed for end users, including online modules, instructor-led sessions, and a sandbox environment for hands-on practice. The information security team requires that all new systems undergo a penetration test before production deployment, with findings above medium severity remediated prior to go-live approval. Dependencies on third-party services were documented and risk-assessed, with fallback mechanisms defined for each critical integration point. The solution was designed to be modular, allowing individual components to be upgraded or replaced without affecting the overall system availability or data integrity. The project governance structure includes a steering committee meeting monthly, a working group meeting weekly, and daily stand-ups during active development sprints. The design considers both current-state constraints and target-state aspirations as documented in the strategic roadmap. The enterprise architecture repository in Sparx EA is the authoritative source for all model elements; any changes to this document must be reflected there within five business days.

The technology vendor recommendations has been a critical area of focus for ArchiSurance since the digital transformation initiative began in 2023. Post-implementation review sessions are scheduled for thirty and ninety days after go-live to capture lessons learned and identify optimization opportunities. Deployment windows are scheduled every Tuesday and Thursday between 06:00 and 08:00 CET, with emergency deployments requiring approval from both the IT Service Manager and the on-call engineer. An internal audit in March 2025 confirmed that the controls implemented for this area meet the requirements of the enterprise risk management framework. The business process documentation follows the BPMN 2.0 standard, with process models maintained in the Signavio process modeling tool and linked to the ArchiMate model elements in Sparx EA. The solution was designed to be modular, allowing individual components to be upgraded or replaced without affecting the overall system availability or data integrity. Cross-functional workshops held in January and March 2025 helped refine the scope and integration points. The total cost of ownership analysis, prepared by the Finance Department in collaboration with IT, indicated a payback period of approximately eighteen months assuming current utilization rates. Automated deployment pipelines leverage the CI/CD infrastructure described in the platform engineering documentation. Budget allocation was approved as part of the annual IT investment portfolio review conducted in November 2024.

Business continuity considerations were incorporated from the outset, with the DR Recovery Node in the secondary data center providing failover capacity; the most recent failover test was conducted in October 2024 and achieved recovery within the target RTO of one hour. Risk mitigation strategies include regular checkpoint reviews, automated regression testing, and staged rollout procedures. Automated deployment pipelines leverage the CI/CD infrastructure described in the platform engineering documentation. The incident management process defines four severity levels, with Severity 1 incidents requiring a response within fifteen minutes and a post-incident review within five business days. The vendor risk assessment framework evaluates suppliers across five dimensions: financial stability, technology roadmap alignment, support quality, security posture, and contractual flexibility. The enterprise architecture repository in Sparx EA is the authoritative source for all model elements; any changes to this document must be reflected there within five business days. Collaboration with the vendor support team has been essential during the implementation phase; weekly technical syncs ensure that configuration changes are validated against the vendor's reference architecture before being applied to the production environment. Lessons learned from the pilot phase have been incorporated into the current design iteration. Cross-functional workshops held in January and March 2025 helped refine the scope and integration points. The initiative is tracked in the enterprise PMO tool with bi-weekly status updates to the steering committee.

Technical debt in this area was cataloged during a dedicated sprint in Q3 2024; the resulting backlog of forty-seven items has been prioritized by business impact and is being addressed alongside feature development. A proof-of-concept was conducted over a four-week period, involving representatives from the Claims Department, IT Operations, and the Data Analytics Team to validate the technical feasibility. During the initial assessment phase, the team identified several areas where the existing architecture fell short of the target-state requirements defined in the enterprise roadmap. The platform engineering team provides a standardized deployment template that all application teams are required to use, ensuring consistency in logging, health checks, and resource limits. The vendor risk assessment framework evaluates suppliers across five dimensions: financial stability, technology roadmap alignment, support quality, security posture, and contractual flexibility. The business process documentation follows the BPMN 2.0

standard, with process models maintained in the Signavio process modeling tool and linked to the ArchiMate model elements in Sparx EA. The incident management process defines four severity levels, with Severity 1 incidents requiring a response within fifteen minutes and a post-incident review within five business days. The observability strategy follows the three pillars model: structured logs shipped to Elasticsearch, distributed traces collected via OpenTelemetry, and metrics scraped by Prometheus; all three are unified in Grafana dashboards. Compliance with the General Data Protection Regulation requires that personally identifiable information is encrypted at rest and in transit, with access restricted to authorized roles as defined in the Identity and Access Management policy. Risk mitigation strategies include regular checkpoint reviews, automated regression testing, and staged rollout procedures.

7. Appendices

References and Related Documents

Doc ID	Title	Version
EA-OVR-001	Enterprise Architecture Overview	3.0
EA-APP-002	Application Portfolio & Inventory	4.2
EA-INFRA-003	Infrastructure Design Document	2.5
EA-BIZ-004	CRM Business Process Description	2.0
EA-CLM-005	Claims Processing Workflow	3.1
EA-SEC-006	Security Architecture	2.3
EA-INT-007	Integration Architecture	2.8
EA-DAT-008	Data Architecture & Governance	2.0
EA-CLD-009	Cloud Migration Assessment	1.5
EA-DR-010	Disaster Recovery & Business Contin	3.0
EA-NET-011	Network Architecture	2.2
EA-ROAD-012	Strategic IT Roadmap 2025-2027	2.0
EA-SVC-013	IT Service Catalog	3.5
EA-VND-014	Vendor & Technology Assessment	1.8
EA-OPS-015	Operational Runbook: Core Platform	4.1
EA-CMP-016	Compliance & Regulatory Requirement	2.5
EA-CHG-017	Change Management: ERP Migration	1.2

Distribution List

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J. van den Berg	Enterprise Architect	Review & Approve
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T. Mulder	Process Architect	Inform
R. de Jong	CISO	Review & Approve
L. Hendriks	Data Architect	Inform
A. Willems	CTO	Final Approval
B. Smit	Head of Claims	Business Sign-off

Detailed Revision Notes for Version 2.1

Section 2.3 was updated to reflect the new naming convention adopted by the Platform Engineering team. All references to 'microservice cluster' have been replaced with 'application workload group' per the Architecture Board decision of January 2025. Section 4.1 was expanded to include the results of the capacity planning exercise conducted in Q4 2024, which identified the need for additional compute resources in the Amsterdam primary data center.

Appendix B was added to include the RACI matrix requested by the PMO during the Q3 2024 quarterly review. The glossary was updated to include twelve new acronyms that were introduced since the previous version. Figure 3 was replaced with an updated diagram generated from the Sparx EA repository on March 28, 2025. The document classification was elevated from 'Internal' to 'Internal - Confidential' following the data classification review of February 2025.

Feedback from the peer review conducted by M. de Vries on March 10, 2025, addressed three items: (1) clarification of the boundary between the DMZ and the application tier, (2) addition of the backup retention policy reference, and (3)

correction of the SLA target for the Claims Handling Service from 99.5% to 99.9% as agreed in the contract amendment of December 2024.

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